AEROSPACE & DEFENSE

For defense tech startups, red tape is a burden—and also a moat

By Andrew Woodman

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At the start of summer, the world witnessed a watershed moment in modern warfare. In its boldest operation yet, the Ukrainian military launched a coordinated attack on five Russian airbases. In a single day, drones costing just a few hundred dollars each inflicted damage estimated at $2.5 billion.

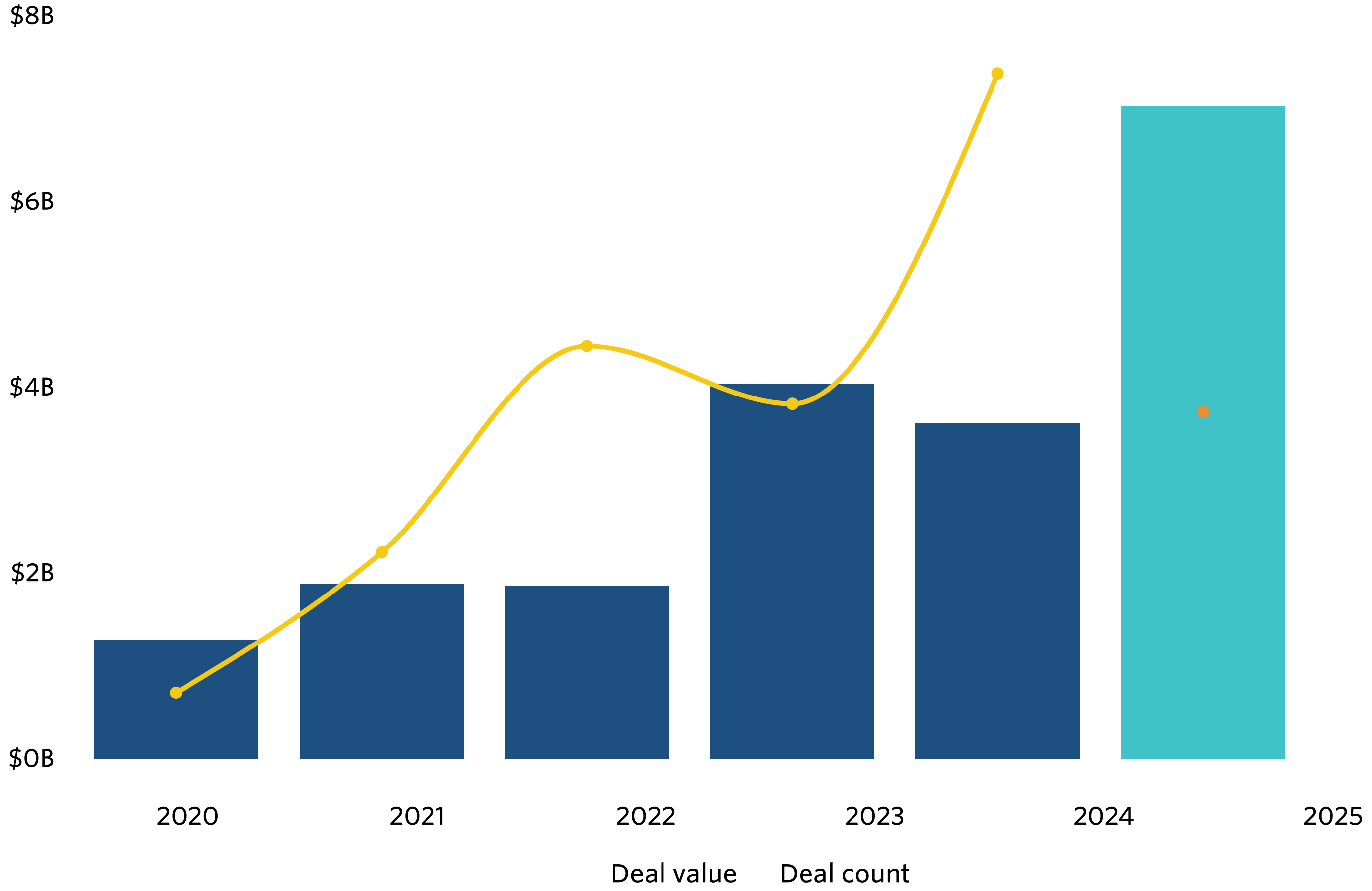
It wasn't just a display of tactical audacity, but also a demonstration of how tech is increasingly driving the narrative of conflict in the 21st century.

"The face of war has changed. It's now about electronic warfare, drone warfare, intelligence—so that all of the existing infrastructure we've got around defense now needs to be completely updated," said Mark Boggett, CEO of London-based VC Seraphim Space. "This is a huge opportunity that the dual-use and defense sector is right in the center of."

New global conflicts fuel VC investment into defense and dual-use technologies, which have both civilian and military applications. At the same time, there is a tension between the speed of innovation needed and the pace at which this technology can get into the hands of the people who need it. The most successful defense tech startups will be those able to navigate a complex maze of regulations and compliance to deliver their technology.

PitchBook data shows that the total value of VC investment in defense tech this year has already reached a new peak, with just over $7 billion invested across 112 rounds globally. This is just over four times the level of investment seen in 2022, when Russia's full-scale invasion of Ukraine began. Much of the investment has been driven by advances in AI and robotics.

VC defense tech investment has soared this year



Source: PitchBook • As of Aug. 5, 2025

The year's largest round was autonomous weapons developer Anduril's $2.5 billion in June, which valued the company at $30.5 billion. Another notable round was a €600 million (about $683 million) investment in Germany-based Helsing, which develops strike drones and underwater surveillance systems. The investment valued the company at €11.4 billion.

The private sector is now the primary driver for rapid defense innovation, with startups iterating and deploying new technologies within a shorter timeframe than traditional government R&D. This change has resulted in a shift in the role of governments, from being inventors to adopters and enablers

"Whereas previously, the R&D was done by defense itself, or by a small number of trusted primes, now it's being done by the broader entrepreneurial marketplace," said Seraphim's Boggett. "What governments are doing is making clear to the market what their needs [and] requirements are, and then they are allowing the market to go ahead and develop them."

The advantage of the private sector stepping in is that militaries can potentially access more cutting-edge technologies faster. Still, it also pressures procurement systems to adapt and keep up with the pace of commercial innovation.

Startups and their defense clients face the challenge of getting new technology to those who need it on time. A common complaint is that government procurement cycles are too slow.

Edward Ebbern is a founding partner at MD One, a national security VC firm based in London, and the founder of the British Army's Defence BattleLab tech accelerator. He explains that the traditional defense procurement system was designed for big capital programs with long timelines.

"The procurement timelines still take years, rather than weeks and months, whereas software is iterated in what feels like hours sometimes, and then you've got entirely new technologies like AI coming in," Ebbern said.

Valley of death

The mismatch between defense acquisition time and the pace of change isn't just a bureaucratic headache. It's often the difference between a breakthrough making it to the field or a startup folding before it reaches production.

"The biggest hurdle for defense tech startups, in my opinion, is going from prototype contract to production contract in a timeline fast enough for a VC cycle," said JD Englehart, a London-based senior investment director at In-Q-Tel, a not-for-profit strategic investor that serves as a bridge between the the private sector and the US national security community. "Venture investors are impatient people; you want to see increased conviction in the thesis [and] increased traction, and you have to show something for that next funding round."

The system exists partly because governments face a complex risk-reward calculation when investing in defense technology. They must balance the need for innovation against the inherent risks of backing unproven technologies. Englehart added that one of the most systemic issues is a culture of risk aversion, especially when public money is at stake and the consequences of failure are dire.

The result is a procurement landscape that favors mature, proven systems, which could lengthen the time it takes startups to have their technology approved for government use. However, not all governments operate in the same way. Some procurement systems are geared toward accommodating faster technology iterations.

Nicholas Nelson, a UK-based general partner with European defense investor Archangel Ventures, noted that the US also benefits from more established pathways for rapid procurement and iteration, adding that the market is generally further ahead in accommodating defense tech startups.

This contrasts with Europe, where procurement pathways vary widely across the region. So, for example, pathways in Estonia, where Archangel invests, bear little resemblance to how the system works in Spain or Italy. As one might expect, countries like Poland and those in the Baltics and Nordics are under greater pressure to move faster due to their proximity to Russia and Ukraine.

Governments are making efforts to slash red tape to help speed up adoption. In June, the US government introduced the Speed Act, a bipartisan initiative to cut bureaucracy in defense procurement.

> Once you get in, it's really hard to dislodge incumbents, especially if you're performing well
>
> Nicholas Nelson, Archangel Ventures

The EU, meanwhile, is introducing a fast-track permitting regime. Member states are required to have a single point of contact for defense companies to submit permit requests, with authorities urged to respond within 60 days. This is also happening on a national level in Europe. Last month, the German government introduced a law to speed up defense procurement by simplifying legal procedures and encouraging European cooperation.

Nevertheless, given that the timelines can still pressure defense tech startups, pursuing a dual-use can offer many startups a way to generate the income needed to keep operating and fund further development.

According to Anmol Manohar, co-founder of propulsion systems supplier and MD One portfolio company Greenjets, pursuing a dual-use strategy comes with trade-offs.

"If you're a purely civilian use case, you can tap into global supply chains quite easily, and therefore, you become very price-competitive in the market," he said. "You can't do that as a dual-use because you're still catering to defense applications. So, you can't tap into any supply chain anywhere. You need to have that sovereign angle."

These restrictions are not limited to the supply chain, though. Dual-use products, even if mostly sold to civilian customers, are also often subject to strict export controls if they have any defense applications.

Seraphim's Boggett also notes that investors in dual-use technology must consider a startup's cap table, which can be a significant stumbling block when the time comes to exit. This is especially true when considering that the world's most liquid M&A market is in the US.

"Companies tend to make sure that they're whiter than white when it comes to dealing with the right countries and the right customers and where they take the money from," he said, adding that defense tech startups must be aware of the Committee on Foreign Investment in the United States. This interagency US government body can review, block or even unwind acquisitions involving US companies if the startup previously took investment from problematic foreign sources.

The sticky, secret sauce

But there is a flipside: the ability to navigate these challenges also gives startups in the space an edge.

"How you source your supply chain in a way that is unique becomes a secret sauce, as much as the technology," explains Manohar of Greenjets. "Being able to source from a supply chain that both customers are happy with, and your price competitiveness becomes an edge in its own right."

Mastering operational complexities becomes a powerful differentiator. It builds trust with customers and embeds a company deeply into long-term programs, creating the kind of entrenched position that competitors struggle to break.

"Once you get into this ecosystem for defense-first companies, it's incredibly sticky," says Archangel's Nelson. "Yes, they're long recurrent timelines, but once you get in, it's really hard to dislodge incumbents, especially if you're performing well."

This is why the defense sector remains attractive for investors, despite logistical challenges and long procurement cycles. Companies that establish themselves and perform well in the ecosystem have the potential to offer long-term, predictable revenue and strong customer loyalty—qualities that are often absent in fast-moving commercial markets.

This can translate into stable cash flow for institutional investors even if growth trajectories are slower. However, this stability comes with its own unique expectations.

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For startups in the defense space, success isn't measured by how quickly a company can scale at all costs, but by how sustainably it can deliver mission-critical capabilities without compromising quality, security, or oversight.

"A true idiosyncrasy to this sector, as opposed to other sectors, is you don't want exponential growth," said William McManners, a managing partner at MD One. "If you get that, we're in real trouble. It requires, in my view, real measure and real maturity in making sure that we support the best products and services getting to where they need to go, but not accidentally creating a flywheel that just goes the wrong way."

This underscores the reality that while procurement processes will likely remain a significant barrier, they are also an important guardrail. As such, shortening those timelines is not always realistic or desirable. MD One's Ebbern states that there are many practical reasons why red tape exists: to prevent poorly informed requirements and ensure proper vetting.

McManners agrees: "If you make this system faster, or reduce bureaucracy, you also create the opportunity for manipulation. You create the opportunity for fewer checks and balances."

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